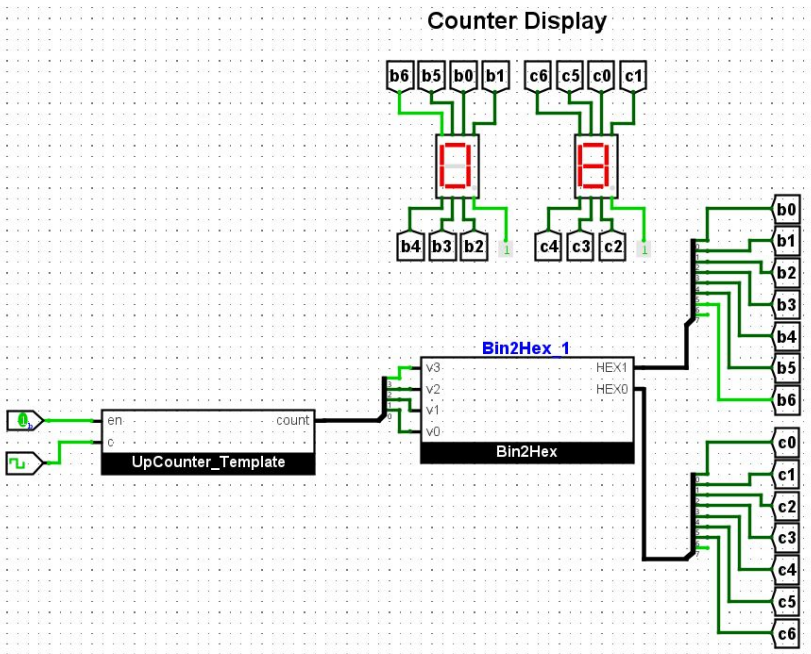
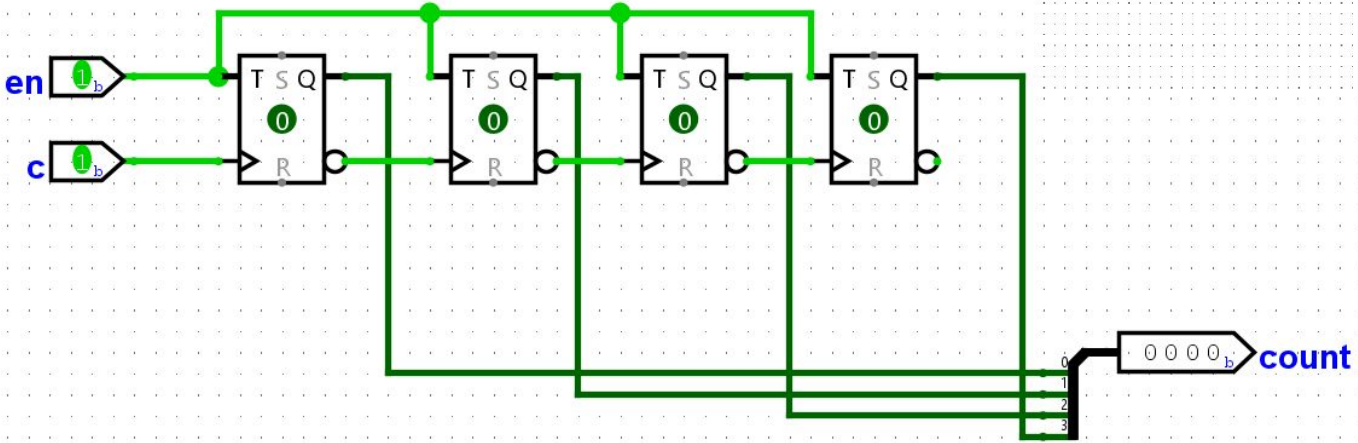
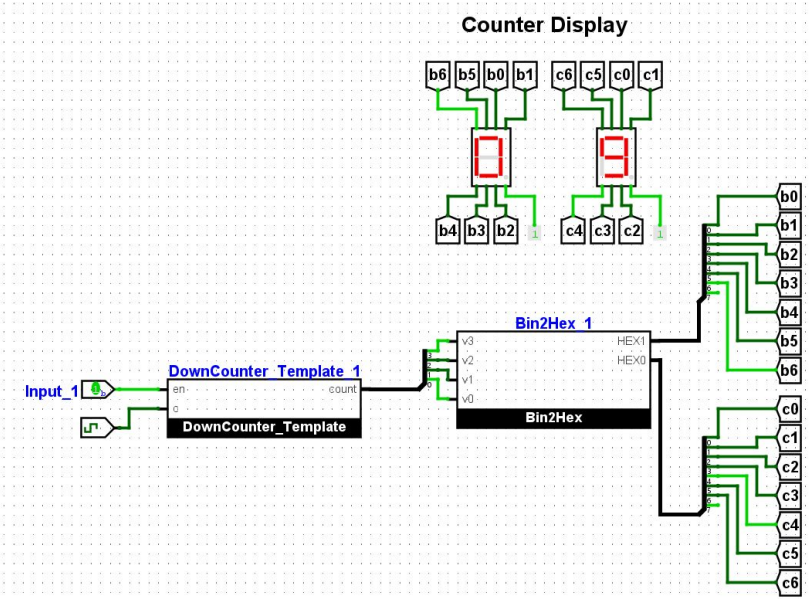
Lab 6: Counters

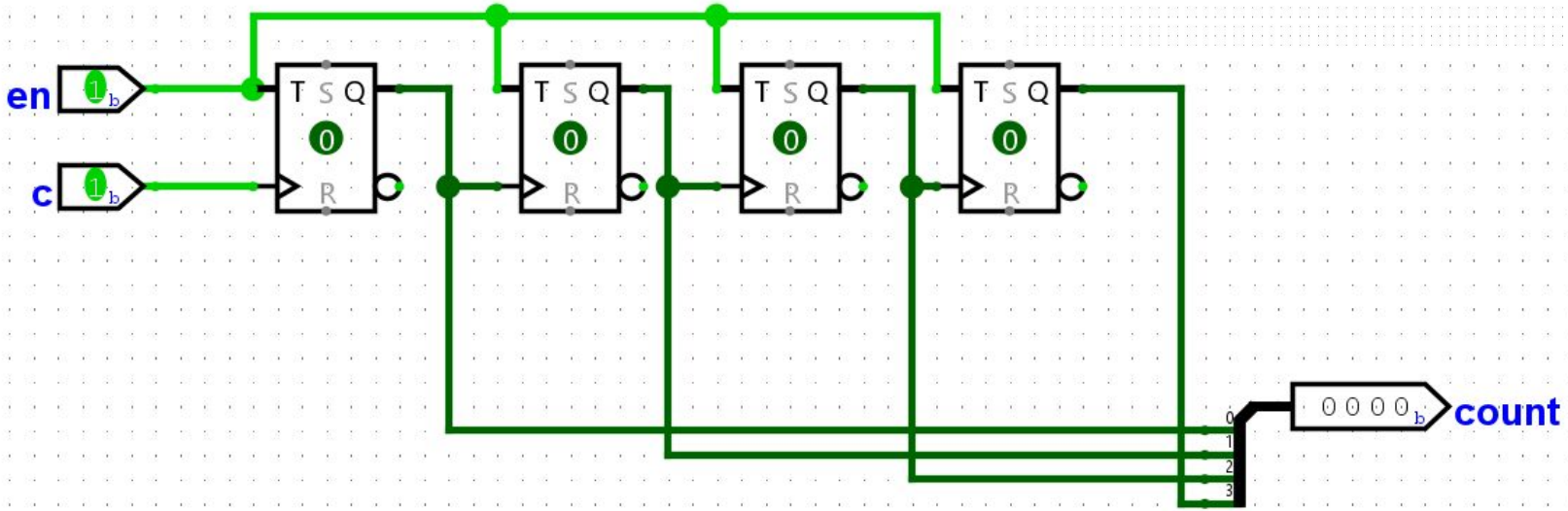
Up counter
(done with class in lab)



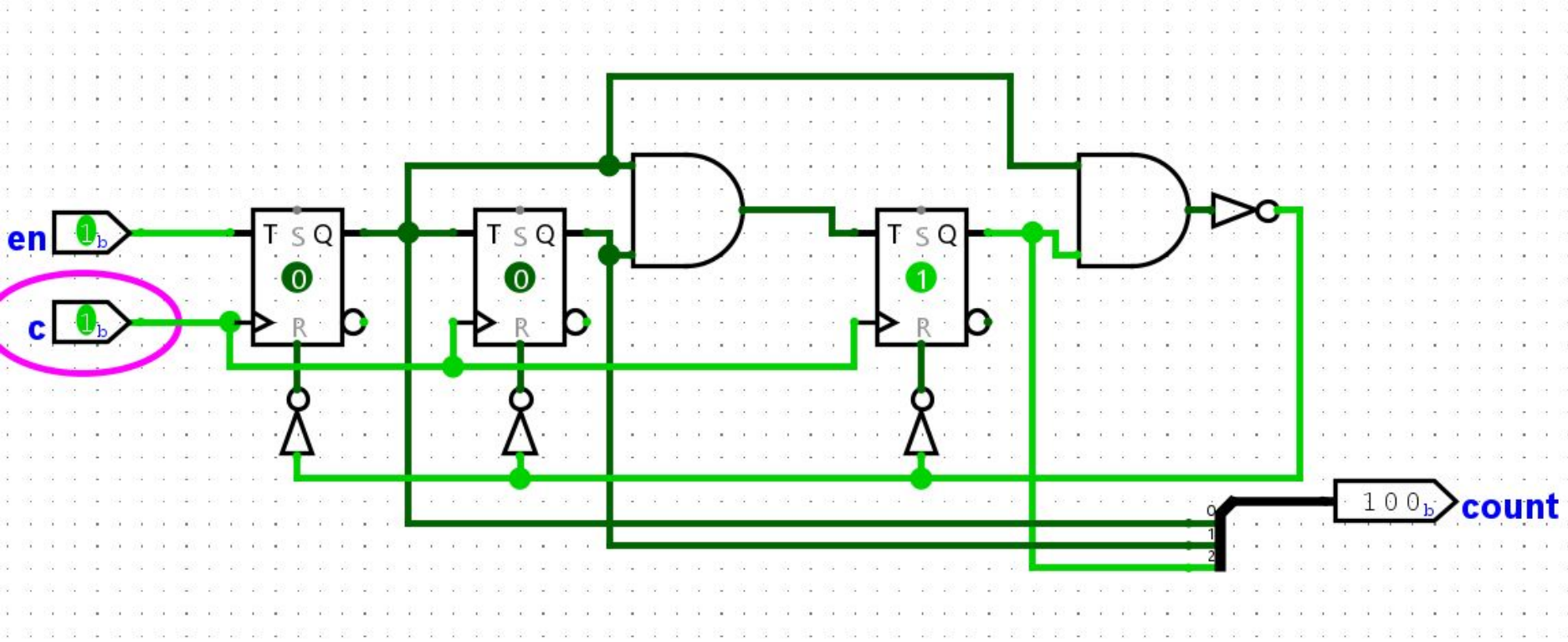


Down counter
(tested and worked!!)

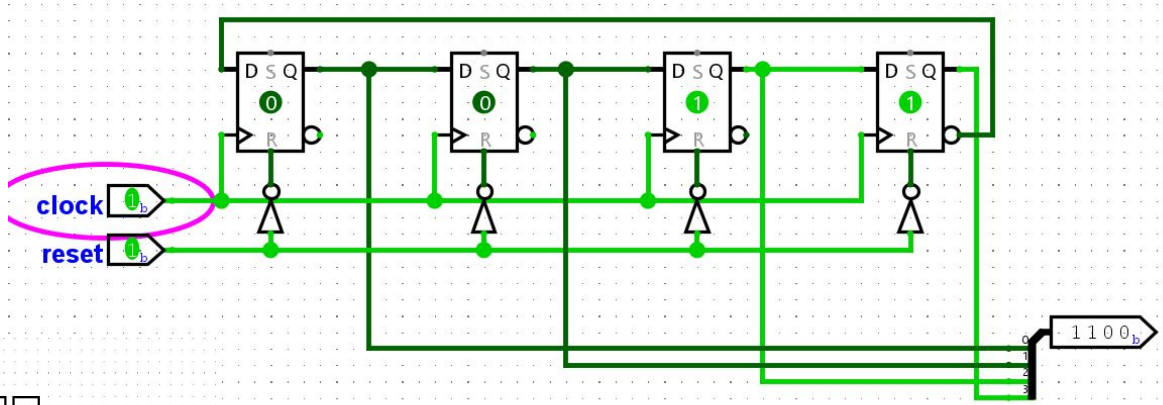


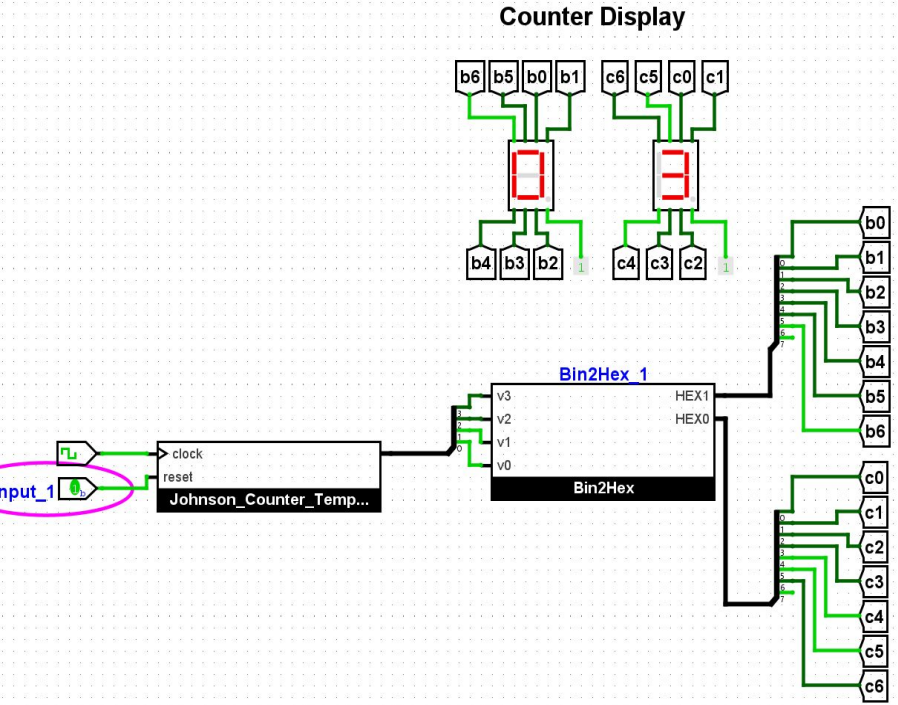


Modulo 5 Counter

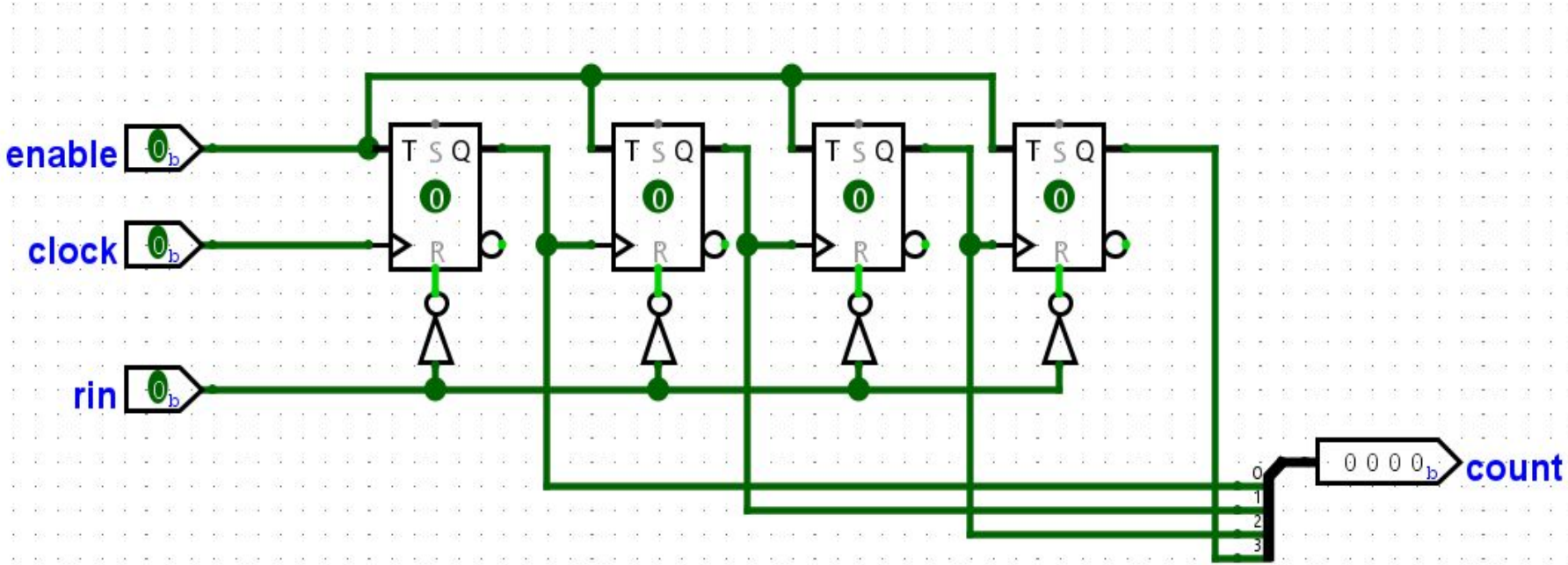


Johnson counter

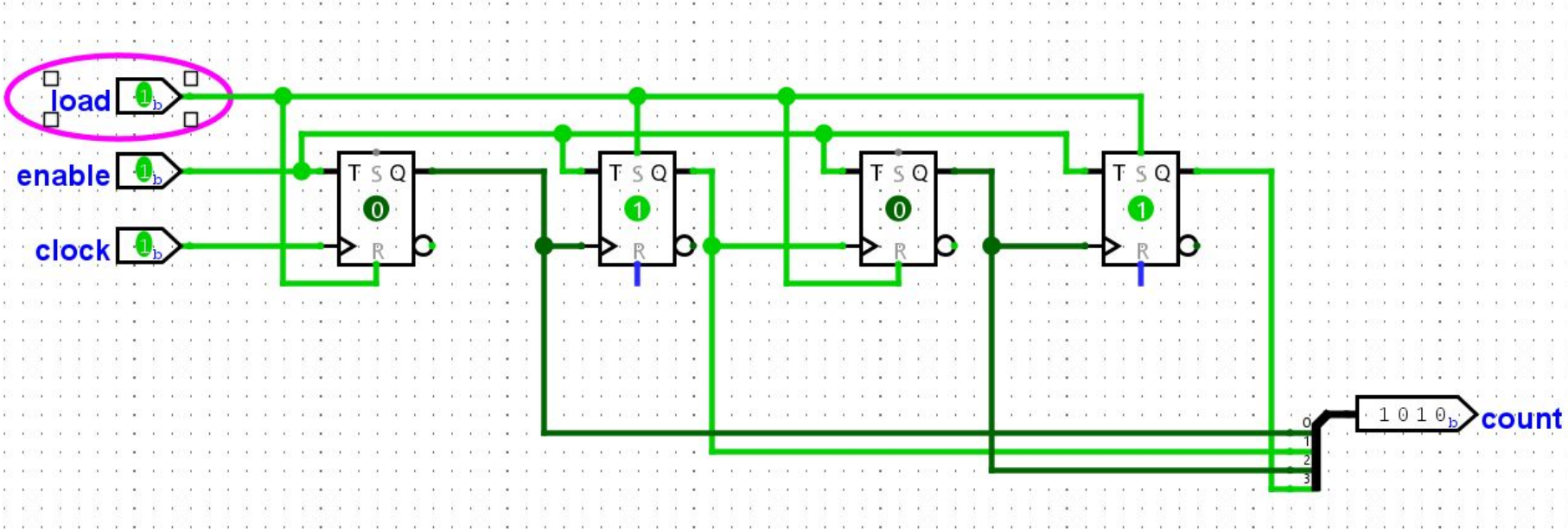




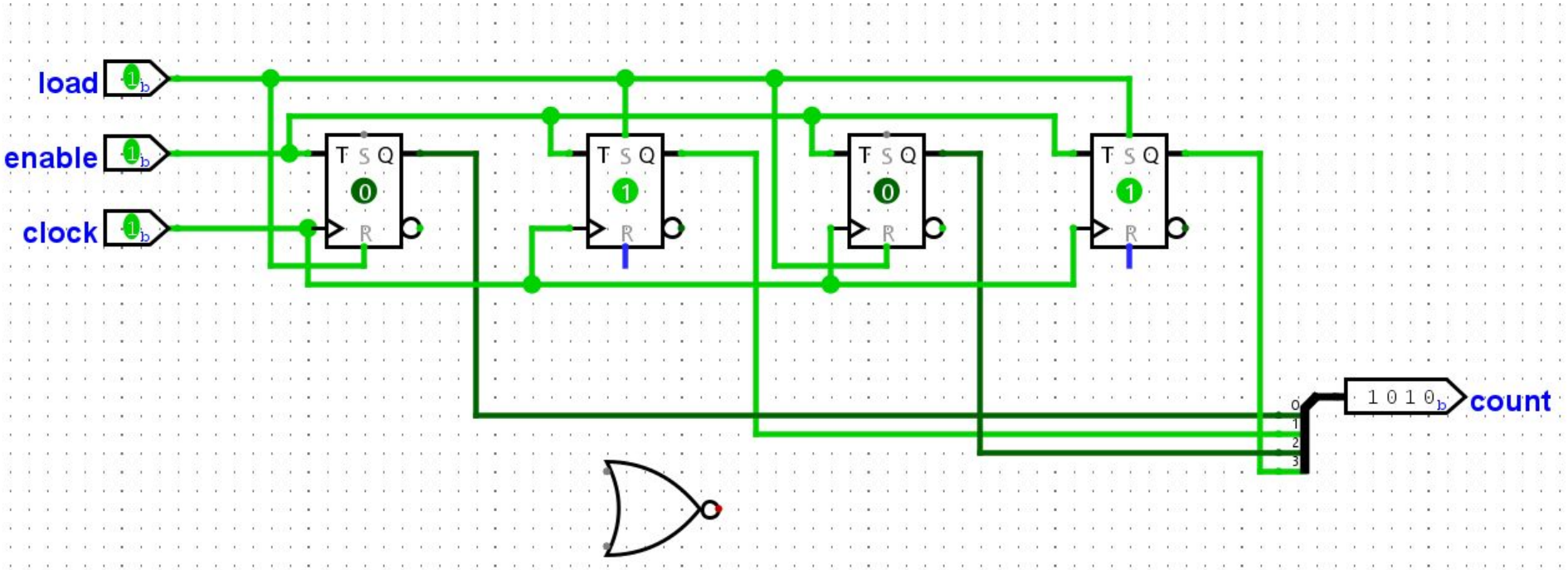
Tried adding enable pin (to down counter)⇒ Resets to 0?
⇒ BAD!!



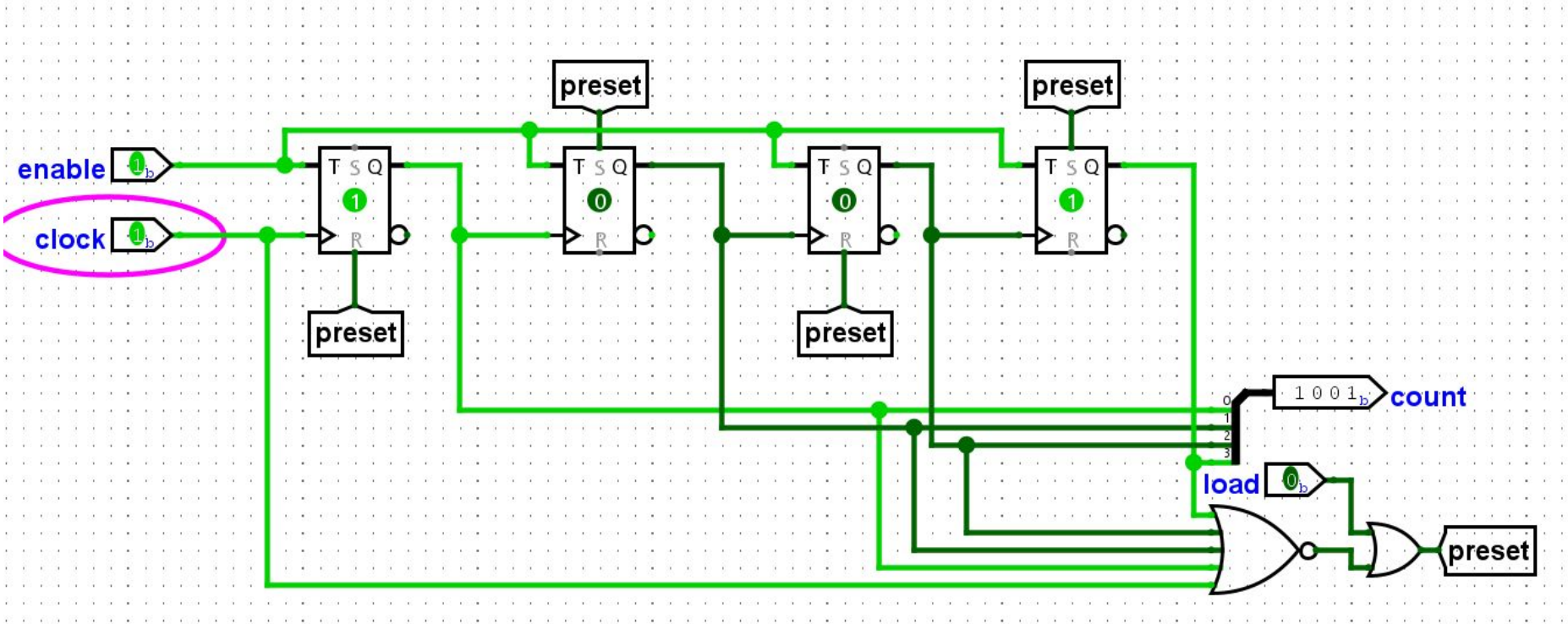
Added a preset of 1010 (ten) to make the count start from ten



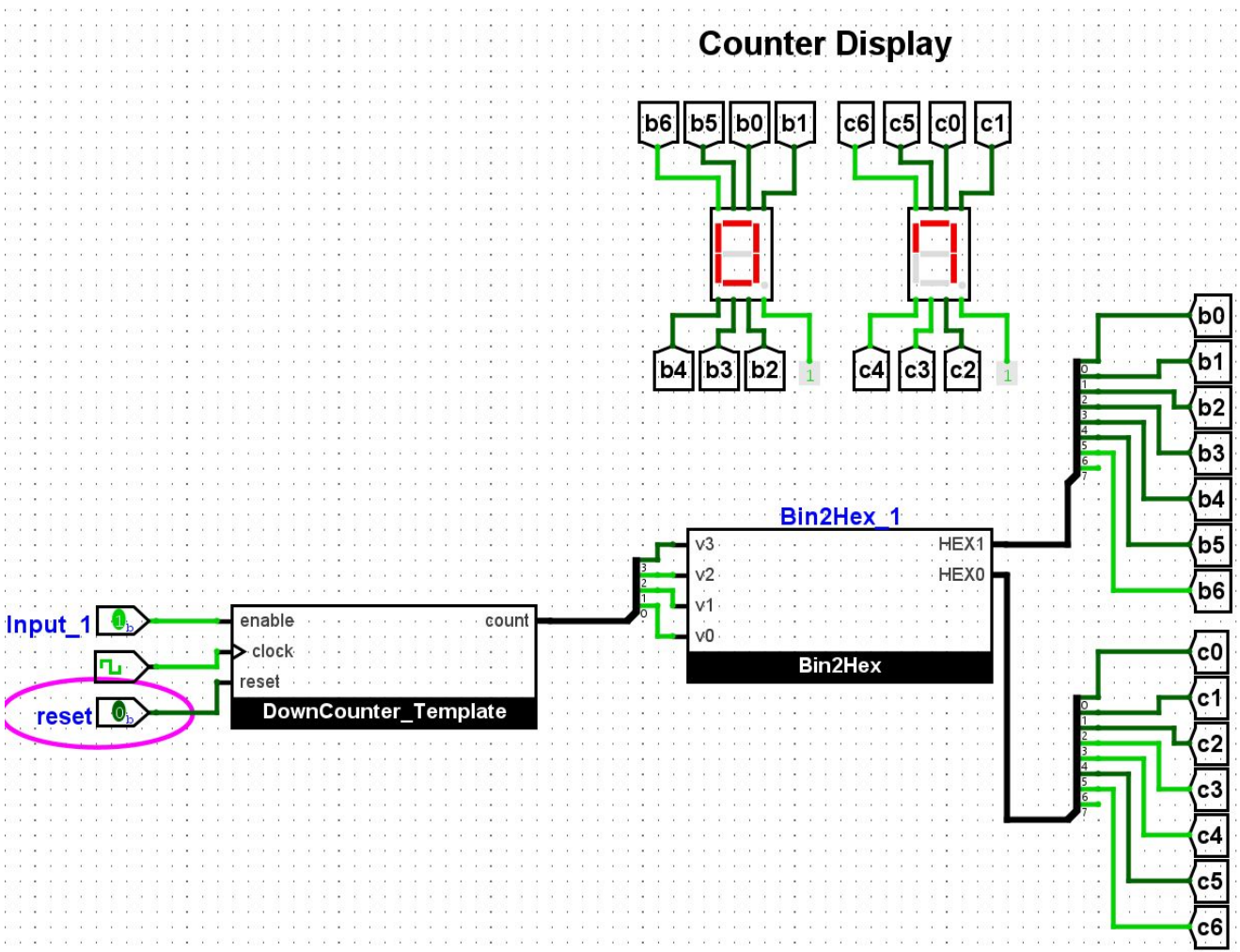
Disconnect the clocks from each Q output (started thinking about using a nor gate to set the condition to restart the count)



Added the nor gate but also connected the clock as an input to include the output 0 in the counter (gemini suggested the clock input and guided my thought process), the or gate let the system either present the load (preset to 10, load works as the reset button) or the output count until it reaches 0000. ⇒ down counter done! button)



Done!!


Counter Display
b6 b5 b0 b1
c6 c5 c0 c1
b4 b3 b2
c4 c3 c2
b0 b1 b2 b3 b4 b5 b6
c0 c1 c2 c3 c4 c5 c6
Bin2Hex_1
v3 v2 v1 v0
HEX1 HEX0
Bin2Hex
Input_1
reset
enable clock reset
count
DownCounter_Template